Closing the loop on aerospace alloys

Autonomous discovery via additive manufacturing

Sterling G. Baird

Vertical Cloud Lab · Mechanical Engineering, BYU

Utah AI Convergence 2026 · Advanced Manufacturing Panel

Sterling Baird, BYU. Ten minutes, one idea: run the alloy-development loop automatically and overnight — the way software teams run automated builds and tests — and the computing tools that make that work transfer to almost any manufacturing sector. Allotment: 2 min on the problem, 4 min on closing the loop, 3 min on the transferable tools, 1 min margin. Acknowledge the organizers and Ashley Spear for the panel invite.

We import the elements our best alloys depend on

- **100%** of manganese — imported
- **100%** of zirconium concentrates — imported
- **74%** of chromium — imported

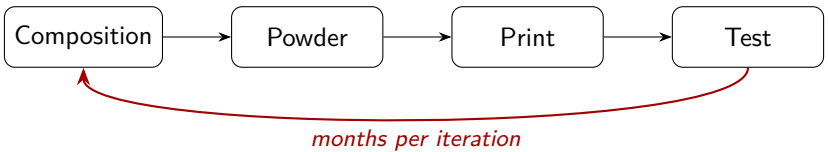
These are the workhorses of high-performance aluminum alloys for additive manufacturing.

The exposure

One supply disruption can idle a domestic AM production line.

Source: USGS 2024. The point isn't the exact numbers — it's that our highest performing structural aluminum alloys are built on elements we don't control. A single supply shock idles the printers. So there are really two goals at once: find high-performing alloys AND reduce how much critical mineral they need. That "dual mandate" is exactly the DOE Genesis Mission Topic 3B framing we're pursuing.

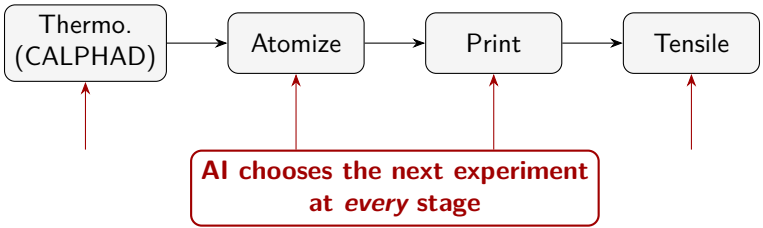
The bottleneck is the development loop, not the printer



Each turn of the loop is slow, manual, and run *one alloy at a time*.

Traditional alloy development is a hand-cranked loop: pick a composition, make powder, print specimens, test them, then a human decides what to try next — and that cycle is months long, run sequentially. That's why a new aerospace alloy can take a decade-plus and millions of dollars. The expensive resource isn't compute, it's wall-clock iterations. Everything that follows is about spending fewer, smarter iterations.

AI can drive the whole pipeline, stage by stage



Model each stage as a node; optimize the workflow end-to-end, cost-aware.

Instead of optimizing the final property in isolation, we model each manufacturing stage — thermodynamics, atomization, printing, testing — as a node in a function network (a directed acyclic graph). The method is Bayesian Optimization over Function Networks, BOFN, with a partial-knowledge-gradient acquisition function. In plain terms: the AI can stop a bad candidate after a cheap CALPHAD check instead of paying to print and test it. It picks the most informative experiment at each stage, accounting for how much that experiment costs. Don't read the acronyms aloud — just say "it decides, stage by stage, what's worth doing next."

Flaime: same loop, run automatically overnight

Screen in silico, in parallel

- $\sim$23,000 element combinations
- thermodynamics + solidification, run in parallel

Validate the few that survive

- top 3–6 systems, 100–200 compositions
- powder $\rightarrow$ print $\rightarrow$ tensile

Phase I target (not yet measured)

Aim: $\sim\mathbf{10}\times$ faster powder-to-property
(planning estimate: $\sim 3\times$ smarter selection $\times$ $\sim 3\times$ closed-loop execution)

Phase I goal

Concept $\rightarrow$ lab-validated closed loop.

This is the project, FLAIME, "Fast Learning AI for Mineral Efficiency." The shape: explore tens of thousands of compositions cheaply and in parallel on HPC, then spend real powder and printer time only on the handful that survive. Roughly 3x from AI choosing better experiments, times another 3x from running the loop closed instead of by hand, gives an order-of-magnitude target. Numbers are targets, not yet results — Phase I is about validating the closed loop on real aluminum alloys, moving from concept (TRL 2) to lab-validated (TRL 4). Software-orchestrated with human operators now; architected for robots next.

These tools transfer to any staged manufacturing workflow

Start optimization without custom code

HONEGUMI generates a runnable optimization template from a few toggles.

Use cheap screens before expensive tests

Stage-aware optimization kills bad candidates early.

Prototype automation on cheap stand-ins

Frugal twins de-risk capital purchases before you buy the expensive machine.

Make experiments shareable and reproducible

Cloud execution plus open, reusable (FAIR) data and models.

This is the slide the organizers asked for — what's useful outside alloys. (1) Honegumi: our open template engine, "skeletal framework" in Japanese — pip install honegumi — it generates a correct, runnable Bayesian-optimization script from a click grid, so a process engineer doesn't have to be a BO expert. (2) The function-network framing: any multi-step process — semiconductors, composites, food, pharma — can be modeled stage-by-stage and optimized cost-aware. (3) Frugal twins: cheap hardware twins to prototype automation at low risk before capital spend. (4) Cloud experimentation plus FAIR data: remote access and shareable data/models. The takeaway: these are domain-agnostic. Point interested attendees to the GitHub organization and Honegumi docs.

Run the loop like software: automate it, share it.

Sterling Baird · github.com/vertical-cloud-lab

Honegumi: honegumi.readthedocs.io · pip install honegumi

Close on the one-liner: treat the manufacturing development loop like a software pipeline — automate the iterations, share the data and the tooling. Point to the lab org and Honegumi. Then hand back to the moderator / open for panel questions. Watch the clock: if short on time, cut the FLAIME numbers slide detail, not the transferable-tools slide.